



Chapter 9

Graphs

MAD101

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Introduction

1 Graphs and Graph Models

Relationship between

- Computers (in a computer network,...)
- Cities (paths, google map,...)
- Webpages (on internet)
- People (in society)
- ...



Introduction

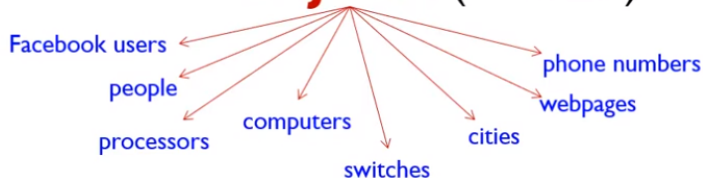
1 Graphs and Graph Models

Represent

relations (edges)

between

objects (vertices)

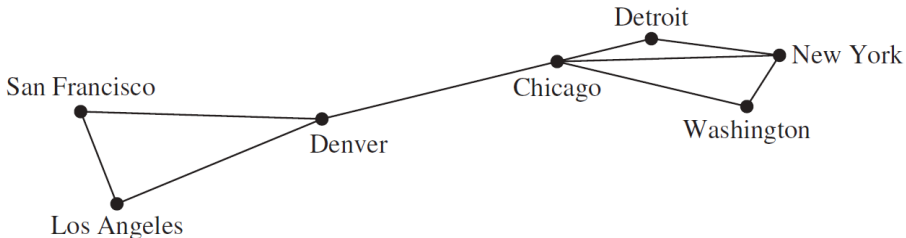


Graph

1 Graphs and Graph Models

Definition 1.1

A graph $G = (V, E)$ consists of V , a nonempty set of **vertices** (đỉnh) (or nodes) and E , a set of **edges** (cạnh). Each edge has either one or two vertices associated with it, called its **endpoints** (đầu mút). *An edge is said to connect its endpoints.*



Remark 1.1. $G = (V, E)$ is called an *infinite graph* if $|V| = +\infty$ or $|E| = +\infty$; otherwise, it is called a *finite graph*.

Some Kind of Graphs

1 Graphs and Graph Models

Definition 1.2

A **directed graph** (or digraph) (V, E) consists of a nonempty set of vertices V and a set of directed edges (or arcs) E . Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to start at u and end at v .

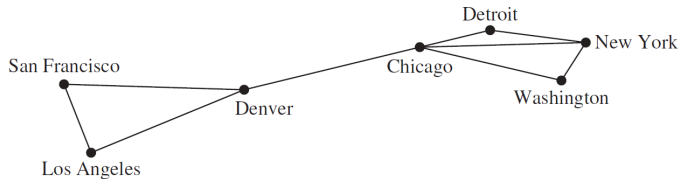
Undirected graph: Each edge has no direction.

<i>Type</i>	<i>Edges</i>	<i>Multiple Edges Allowed?</i>	<i>Loops Allowed?</i>
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

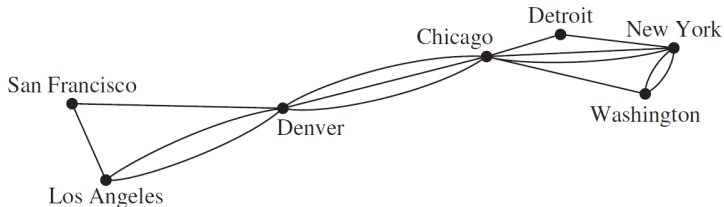
Some Kind of Graphs

1 Graphs and Graph Models

- **Simple graph:** No two edges connect the same pair of vertices



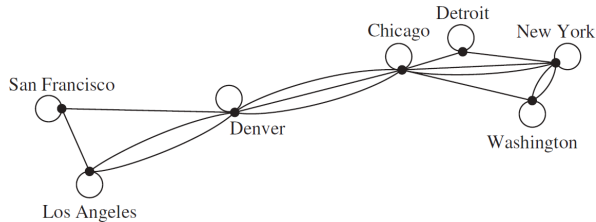
- **Multigraph:** Multiple edges connect the same pair of vertices



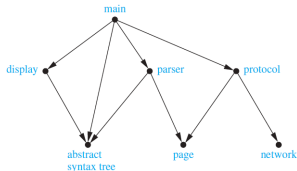
Some Kind of Graphs

1 Graphs and Graph Models

- **Pseudograph:** Multigraph may have **loops** (khuyên, vòng)



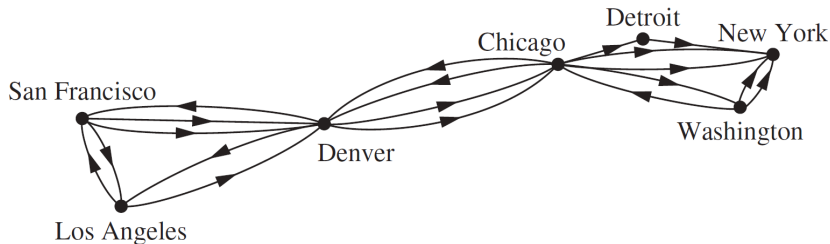
- **Simple directed graph:** has no loops and has no multiple directed edges



Some Kind of Graphs

1 Graphs and Graph Models

- **Directed multigraphs** (or multiple directed edges): have multiple directed edges from a vertex to a second (possibly the same) vertex.

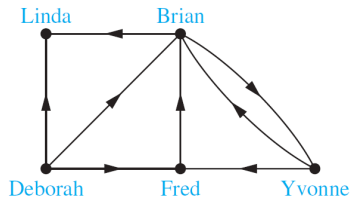


- **Mixed graph:** A graph with both directed and undirected edges.

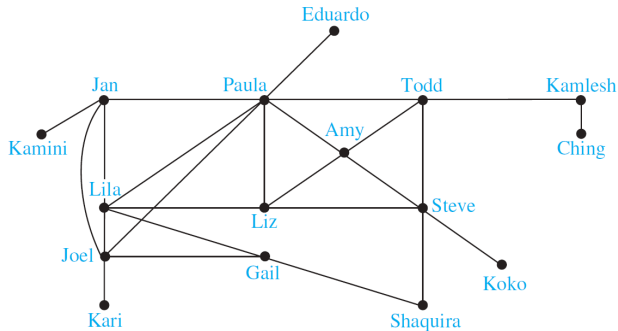
Graph Models

1 Graphs and Graph Models

- An influence graph



- An acquaintanceship graph



Graph Models

1 Graphs and Graph Models

- A precedence graph

S_1 $a := 0$

S_2 $b := 1$

S_3 $c := a + 1$

S_4 $d := b + a$

S_5 $e := d + 1$

S_6 $e := c + d$

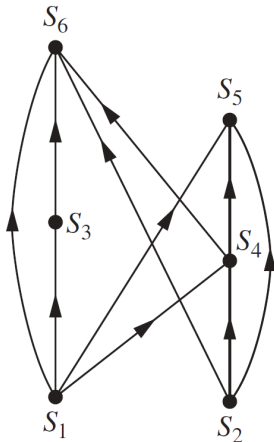




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Basic Terminology

2 Graph Terminology and Special Types of Graphs

Definition 2.1

Two vertices u and v in an undirected graph G are called **adjacent** (or neighbors) (kề nhau) in G if u and v are endpoints of an edge e of G . Such an edge e is called **incident** (cung/cạnh) with the vertices u and v and e is said to connect u and v .

Definition 2.2

The **degree of a vertex** in an undirected graph is the number of edges incident with it, except that a **loop at a vertex contributes twice to the degree of that vertex**. The degree of the vertex v is denoted by **$\deg(v)$** .

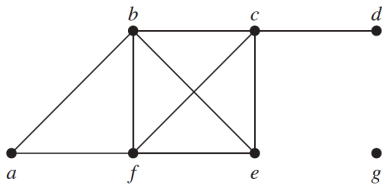
Remark 2.1.

1. A vertex of degree zero is called **isolated** (đỉnh cô lập)
2. A vertex is **pendant** (đỉnh treo) if and only if it has degree one.

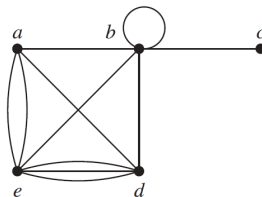
Example

2 Graph Terminology and Special Types of Graphs

Example 2.1. What are the degrees of the vertices in the graphs G and H.



G

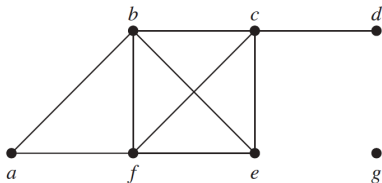


H

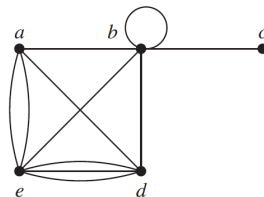
Example

2 Graph Terminology and Special Types of Graphs

Example 2.2. What are the degrees of the vertices in the graphs G and H.



G



H

Solution.

Graph G: $\deg(a) = 2$, $\deg(b) = 4$, $\deg(c) = 4$, $\deg(d) = 1$, $\deg(f) = 4$, $\deg(e) = 3$, $\deg(g) = 0$.

Graph H: $\deg(a) = 4$, $\deg(b) = \deg(e) = 6$, $\deg(c) = 1$, $\deg(d) = 5$.

THE HANDSHAKING THEOREM

2 Graph Terminology and Special Types of Graphs

Theorem 2.1

Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v)$$

(Note that this applies even if multiple edges and loops are present.)

Example 2.3. How many edges are there in a graph with 10 vertices each of degree six?

THE HANDSHAKING THEOREM

2 Graph Terminology and Special Types of Graphs

Theorem 2.3

Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v)$$

(Note that this applies even if multiple edges and loops are present.)

Example 2.4. How many edges are there in a graph with 10 vertices each of degree six?

Solution. $2m = 10 \cdot 6 \implies m = 30$

Theorem 2.4

An undirected graph has an even number of *vertices of odd degree*.



Example

2 Graph Terminology and Special Types of Graphs

Example 2.5. Is there an undirected graph with degree sequence 7, 5, 4, 3, 2, 2.

Example 2.6. How many edges does a graph have if its degree sequence is 5, 5, 4, 3, 2, 1, 0.

Example 2.7. Is there an simple undirected graph with degree sequence 4, 3, 2, 1, 0? How many edges are there? Draw a such graph (if exist).



Example

2 Graph Terminology and Special Types of Graphs

Example 2.8. Is there an undirected graph with degree sequence 7, 5, 4, 3, 2, 2.

Example 2.9. How many edges does a graph have if its degree sequence is 5, 5, 4, 3, 2, 1, 0.

Example 2.10. Is there an simple undirected graph with degree sequence 4, 3, 2, 1, 0? How many edges are there? Draw a such graph (if exist).

Solution.

1. No. 3 vertices of odd degree: 7, 5, 3.
2. 10 edges.
3. Yes, 5 edges.

In-degree & Out-degree

2 Graph Terminology and Special Types of Graphs

Definition 2.3

When (u, v) is an edge of the graph G with directed edges, u is said to be adjacent to v and v is said to be adjacent from u .

$$a \longrightarrow b$$

- a is **initial vertex**
- b is **terminal vertex** (or end vertex)

The initial vertex and terminal vertex of **a loop** are the same.

- **In-degree** ($\deg^-(v)$): the number of edges with v as their terminal vertex,
- **Out-degree** ($\deg^+(v)$): the number of edges with v as their initial vertex.

Remark 2.2. A loop v have $\deg^-(v) = \deg^+(v) = 1$.

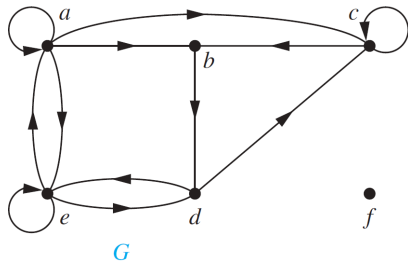
Theorem 2.5

Let $G = (V, E)$ be a graph with directed edges. Then $\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$

In-degree & Out-degree - Example

2 Graph Terminology and Special Types of Graphs

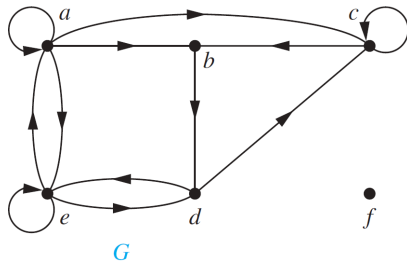
Example 2.11. Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure.



In-degree & Out-degree - Example

2 Graph Terminology and Special Types of Graphs

Example 2.12. Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure.



Solution.

- $\deg^-(a) = 2, \deg^+(a) = 4$
- $\deg^-(b) = 2, \deg^+(b) = 1$
- $\deg^-(c) = 3, \deg^+(c) = 2$
- $\deg^-(d) = 2, \deg^+(d) = 2$
- $\deg^-(e) = 3, \deg^+(e) = 3$
- $\deg^-(f) = \deg^+(f) = 0$



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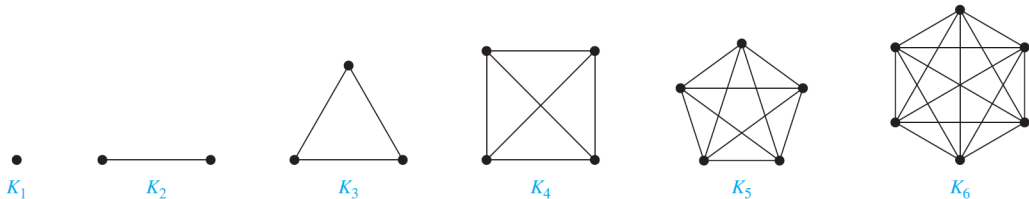
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Complete Graphs

3 Some Special Simple Graphs

Definition 3.1

A **complete graph** on n vertices, denoted by K_n , is a simple graph that contains exactly one edge between each pair of distinct vertices.



- K_n has n vertices and $C_n^2 = \frac{n(n-1)}{2}$ edges. $\left(C_n^k = \frac{n!}{k!(n-k)!} \right)$
- The degree of each vertex is $n - 1$.

Cycles

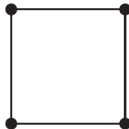
3 Some Special Simple Graphs

Definition 3.2

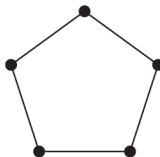
A **cycle** C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.



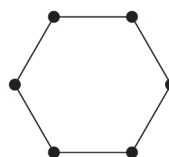
C_3



C_4



C_5



C_6

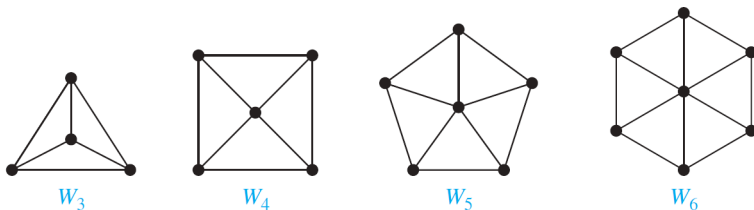
- C_n has n vertices and n edges.
- The degree of each vertex is 2.

Wheels

3 Some Special Simple Graphs

Definition 3.3

We obtain a **wheel** W_n when we add an additional vertex to a **cycle** C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n , by new edges.



- W_n has $n + 1$ vertices and $2n$ edges.
- There are 3-degree of n vertices and **one** vertex has n -degree.

n-Cubes

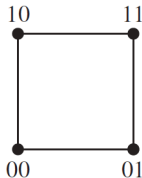
3 Some Special Simple Graphs

Definition 3.4

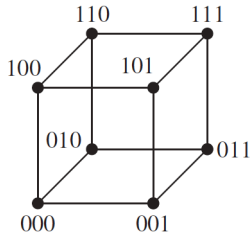
An **n-dimensional hypercube**, or **n-cube**, denoted by Q_n , is a graph that has vertices representing the 2^n bit strings of length n . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.



Q_1



Q_2

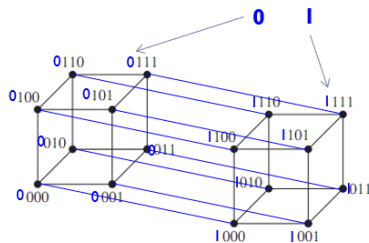


Q_3

n-Cubes

3 Some Special Simple Graphs

Construct Q_4 from two copies of Q_3 .



- Q_n has 2^n vertices and $n2^{n-1}$ edges.
- The degree of each vertex is n .

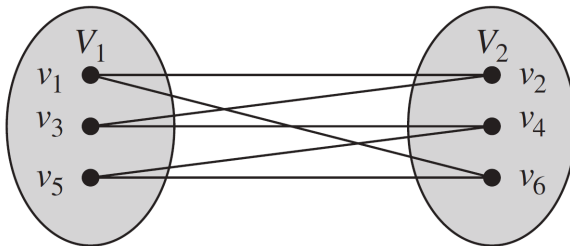
(Mỗi đỉnh nằm trong đồ thị Q_n có bậc là n , tổng số bậc nằm trong đồ thị Q_n là $n2^n$, áp dụng định lý HANDSHAKING ta có kết quả trên.)

Bipartite Graphs

3 Some Special Simple Graphs

Definition 3.5

A simple graph G is called **bipartite** if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a bipartition of the vertex set V of G .



Bipartite Graphs

3 Some Special Simple Graphs

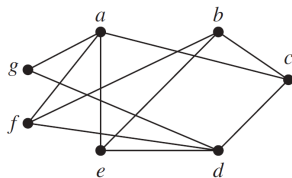
Theorem 3.1

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that **no two adjacent vertices are assigned the same color**.

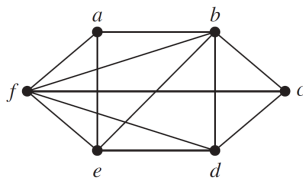
Remark 3.1. How to check?

- Let color the vertices using 2 different colors.
- **Two adjacent vertices must have different colors** (e.g., **red** and **black**).

Example 3.1. Are the graphs G and H displayed in the Figure bipartite?



G



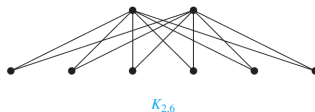
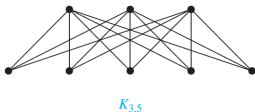
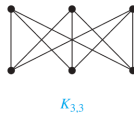
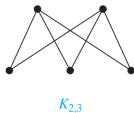
H

Complete Bipartite Graphs

3 Some Special Simple Graphs

Definition 3.6

A **complete bipartite graph** $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.



- $K_{m,n}$ has $m + n$ vertices and mn edges.
- There are n -degree of m vertices and m -degree of n vertices.



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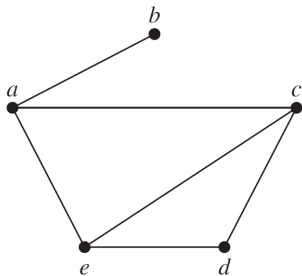
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Adjacency List (Danh sách kề)

4 Representing Graphs & Graph Isomorphism

Adjacency list: For each vertex u in the graph, there is a list of vertex v which there is an edge between u and v .

Example 4.1. Use adjacency lists to describe the simple graph given in Figure.

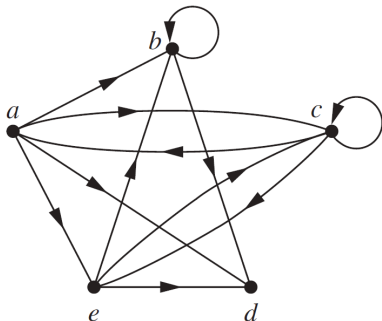


<i>Vertex</i>	<i>Adjacent Vertices</i>
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

Adjacency List (Danh sách kề)

4 Representing Graphs & Graph Isomorphism

Example 4.2. Represent the directed graph shown in Figure by listing all the vertices that are the **terminal vertices** of edges starting at each vertex of the graph.



<i>Initial Vertex</i>	<i>Terminal Vertices</i>
<i>a</i>	<i>b, c, d, e</i>
<i>b</i>	<i>b, d</i>
<i>c</i>	<i>a, c, e</i>
<i>d</i>	
<i>e</i>	<i>b, c, d</i>

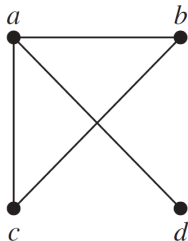
Adjacency Matrices

4 Representing Graphs & Graph Isomorphism

Definition 4.1

Adjacency matrix $A = [a_{ij}]$ with $a_{ij} = \begin{cases} \text{the number of edges that are associated to } \{v_i, v_j\} \\ 0, \text{ Otherwise} \end{cases}$

Example 4.3. Use an adjacency matrix to represent the graph shown in Figure.



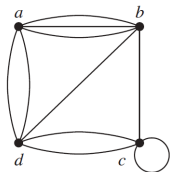
Solution. The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Adjacency Matrices - Example

4 Representing Graphs & Graph Isomorphism

Example 4.4. Use an adjacency matrix to represent the pseudograph shown in Figure.



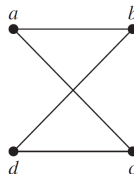
Solution.

$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

Example 4.5. Draw a graph with the adjacency matrix.

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Solution.



Adjacency Matrices - Example

4 Representing Graphs & Graph Isomorphism

Example 4.6. Given the **adjacency matrix** representing an undirected graph G .

1. Find the **deg c**.
2. How many edges are there?

	a	b	c	d
a	1	2	1	0
b	2	0	2	2
c	1	2	1	0
d	0	2	0	0

Solution.

1. $\deg(c) = 1 + 2 + 1.2 + 0 = 5$ (c is loop).
2. 9.

Incidence Matrices

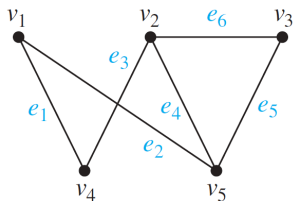
4 Representing Graphs & Graph Isomorphism

Definition 4.2

Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G . Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $M = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1, & \text{when edge } e_j \text{ is incident with } v_i \\ 0, & \text{Otherwise} \end{cases}$$

Example 4.7. Represent the graph shown in Figure with an incidence matrix.



$$\begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Incidence Matrices - Examples

4 Representing Graphs & Graph Isomorphism

Example 4.8. How many column are there in the incidence matrix of W_9 ?

Example 4.9. How many 1s are there in the incidence matrix representing the complete graph K_{10} ?

Example 4.10. How many loops are there? Find $\deg(v_4)$.

$$\begin{array}{c}
 \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{matrix} \\
 \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}
 \end{array}$$

Solution. 18; 90; loops: 2, $\deg(v_4) = 1 + 1.2 = 3$ (loop).

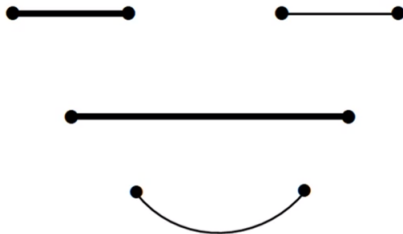
Isomorphism of Graphs (Đồng cấu)

4 Representing Graphs & Graph Isomorphism

Two Simple Graphs are isomorphic.

In this way, they have:

- The same number of vertices.
- The same number of edges.
- The same degrees of the vertices ($\deg(v) = \deg[f(v)]$).
- a and b are adjacent if $f(a)$ and $f(b)$ are adjacent.



THE SAME

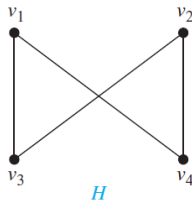
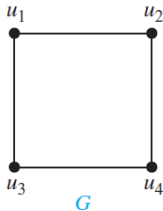
Isomorphism of Graphs

4 Representing Graphs & Graph Isomorphism

Definition 4.3

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a **bijection f** from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 .

Example 4.11. Show that the graphs $G = (V, E)$ and $H = (W, F)$, displayed in Figure, are isomorphic.

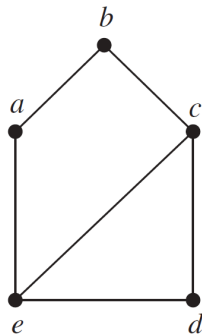


Solution. $f(u_1) = v_1, f(u_2) = v_4, f(u_3) = v_3$, and $f(u_4) = v_2$

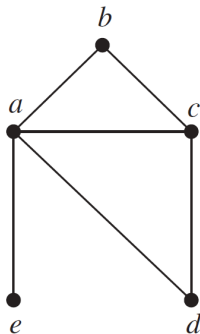
Isomorphism of Graphs - Example

4 Representing Graphs & Graph Isomorphism

Example 4.12. Show that the graphs displayed in Figure are not isomorphic



G

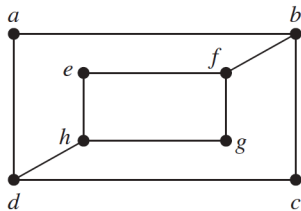


H

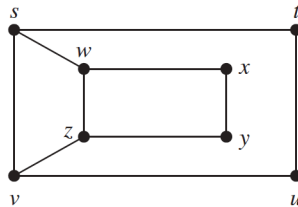
Isomorphism of Graphs - Example

4 Representing Graphs & Graph Isomorphism

Example 4.13. Determine whether the graphs shown in Figure are isomorphic.



G

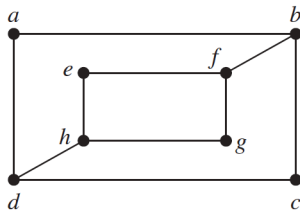


H

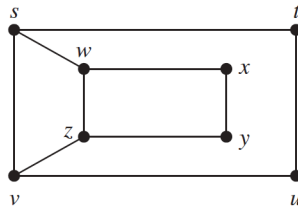
Isomorphism of Graphs - Example

4 Representing Graphs & Graph Isomorphism

Example 4.14. Determine whether the graphs shown in Figure are isomorphic.



G



H

Solution. Not isomorphic because vertex a (2-degree) in G adjacent with b, d (all of them have 3-degree) and in H , there is not exist a vertex which have the same characteristic.



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5 Connectivity

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- ▶ Representing Graphs & Graph Isomorphism
- ▶ **Connectivity**
- ▶ Euler & Hamilton Paths
- ▶ Shortest-Path Problems
- ▶ Preview

Paths & Circuit/Cycle

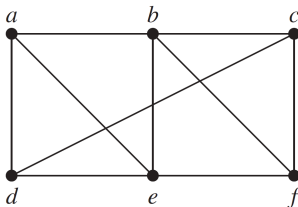
5 Connectivity

Definition 5.1

Let n be a nonnegative integer and G an **undirected graph**.

- A **path** (đường đi) of length n from u to v in G is a sequence of n edges $e_1, \dots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \dots, x_{n-1}, x_n = v$ of vertices such that e_i has, for $i = 1, \dots, n$, the endpoints x_{i-1} and x_i . When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \dots, x_n$.
- The path is a **circuit** (chu trình) if it begins and ends at the same vertex.
- A path or circuit is **simple** (đơn) if it does not contain the same edge more than once.

Example 5.1.



- a, d, c, f, e is a **simple path** of length 4.
- d, e, c, a is not a path.
- b, c, f, e, b is a **circuit** of length 4.

Connectedness In Undirected Graphs

5 Connectivity

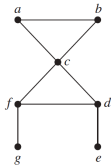
Definition 5.2

An undirected graph is called **connected** (liên thông) if there is a path between every pair of distinct vertices of the graph. An undirected graph that is **not connected** is called **disconnected**.

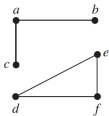
Theorem 5.1

There is a simple path between every pair of distinct vertices of a connected undirected graph.

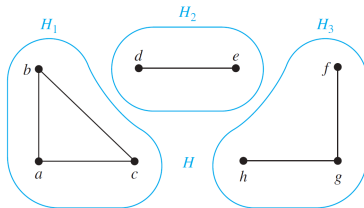
Example 5.2. G_1 is **connected**, G_2 is **disconnected**, H is **disconnected** with 3 connected components (H_1 , H_2 , H_3)



G_1



G_2



H

Strongly connected & Weakly connected

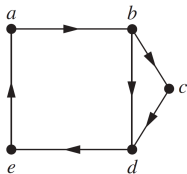
5 Connectivity

Definition 5.3

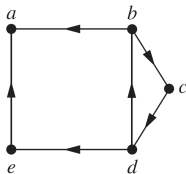
- A directed graph is **strongly connected** if there is a path from a to b and from b to a whenever a and b are vertices in the graph.
- A directed graph is **weakly connected** if and only if there is always a path between two vertices when the directions of the edges are disregarded (không quan tâm) (**underlying undirected**).

Remark 5.1. Any strongly connected directed graph is also weakly connected.

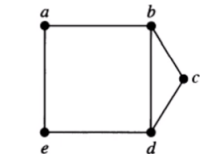
Example 5.3. G is strongly connected, H is weakly connected.



G



H



underlying undirected
graph of H

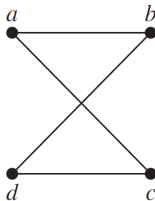
Counting Paths Between Vertices

5 Connectivity

Definition 5.4: *(with directed or undirected edges, with multiple edges and loops allowed)*

- Let G be a graph with **adjacency matrix A** with respect to the ordering $v_1, v_2, \dots, v_n$ of the vertices of the graph,
- The number of different paths of length r from v_i to v_j equals the (i, j) th entry of A^r** , where r is a positive integer.

Example 5.4. How many paths of length four are there from a to d in the simple graph G in figure



Solution of Example 5.4

5 Connectivity

- The adjacency matrix of G (ordering the vertices as a, b, c, d) is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

- Thus, we have

$$\mathbf{A}^4 = \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix}$$

- There are 8 paths of length four from a to d .

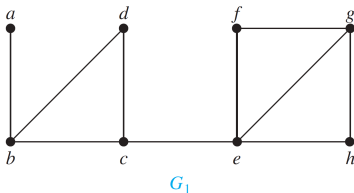
Cut vertices & Cut edges

5 Connectivity

Definition 5.5

- **Cut vertex (or articulation point)** (điểm cắt, đỉnh khớp): the removal from a graph of a vertex and all incident edges produces a subgraph with more connected components.
- **Cut edge (or bridge)** (cạnh cắt, cầu nối): An edge whose removal produces a graph with more connected components than in the original graph.

Example 5.5. Find the cut vertices and cut edges in the graph shown in Figure.

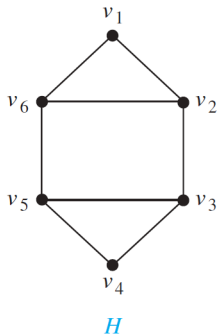
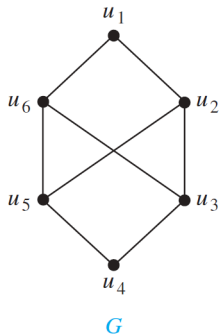


Solution. Cut vertices: b, c, e; Cut edges: $\{a, b\}$ and $\{c, e\}$.

Path and Isomorphism

5 Connectivity

Example 5.6. Using path to determine whether two graphs are isomorphic.



G and H have the same

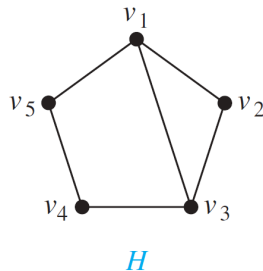
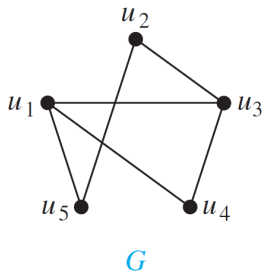
- Number of vertices
- Number of edges
- 2 vertices degree 2
- 4 vertices degree 3

But, H has a simple circuit with length 3 and G has a simple circuit with length 4. Thus, they are not isomorphic.

Path and Isomorphism

5 Connectivity

Example 5.7. Using path to determine whether two graphs are isomorphic



They are isomorphic.



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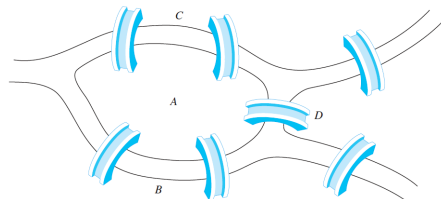
6 Euler & Hamilton Paths

- ▶ Graphs and Graph Models
- ▶ Graph Terminology and Special Types of Graphs
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- ▶ Connectivity
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Introduction

6 Euler & Hamilton Paths

The town of Königsberg, Prussia (now called Kaliningrad and part of the Russian republic), was divided into four sections by the branches of the Pregel River. These four sections included the two regions on the banks of the Pregel, Kneiphof Island, and the region between the two branches of the Pregel. In the eighteenth century seven bridges connected these regions. Figure depicts these regions and bridges. The townspeople took long walks through town on Sundays. They wondered whether it was possible to **start at some location in the town, travel across all the bridges once without crossing any bridge twice, and return to the starting point.**



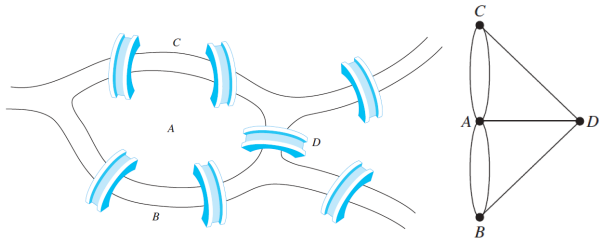
Introduction

6 Euler & Hamilton Paths

The Swiss mathematician Leonhard Euler solved this problem. His solution, published in 1736, may be the first use of graph theory.



Figure: Euler (1707 – 1783)



Can one travel across all the bridges once and return to starting point?
His answer: All of vertices must have **even-degree**.

Euler Paths & Circuits

6 Euler & Hamilton Paths

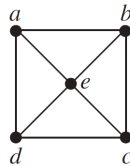
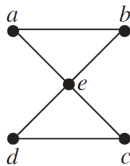
Definition 6.1

- An **Euler circuit** in a graph G is a simple circuit **containing every edge of G** ,
- An **Euler path** in G is a simple path **containing every edge of G** .

Theorem 6.1

1. A connected multigraph with at least two vertices has an **Euler circuit** if and only if each of its vertices has **even degree**.
2. A connected multigraph has an **Euler path** but not an Euler circuit if and only if it has exactly **two** vertices of **odd degree**.

Example 6.1. G_1 : has an Euler circuit, G_2 : has no Euler circuit.



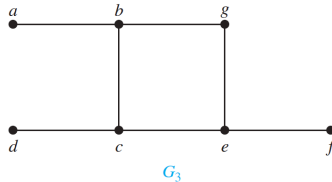
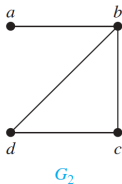
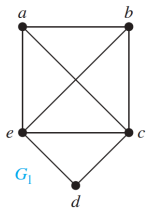
Hamilton Paths & Circuits

6 Euler & Hamilton Paths

Definition 6.2

- A simple path in a graph G that passes **through every vertex** exactly once is called a **Hamilton path**.
- A simple circuit in a graph G that passes **through every vertex** exactly once is called a **Hamilton circuit**.

Example 6.2. G_1 has Hamilton circuit: a, b, c, d, e, a ; G_2 has no Hamilton circuit; G_3 has neither a Hamilton circuit nor a Hamilton path.



Hamilton Circuits - Sufficient conditions

6 Euler & Hamilton Paths

Although no useful necessary and sufficient conditions for the existence of Hamilton circuits are known, quite a few sufficient conditions have been found.

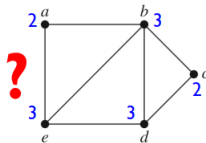
Dirac's theorem.

G is a graph:

- *simple*
- $n (\geq 3)$ vertices
- $\forall v_i, \deg(v_i) \geq \frac{n}{2}$



G has a **Hamilton circuit**



Ore's theorem.

G is a graph:

- *simple*
- $n (\geq 3)$ vertices
- $\forall u, \forall v, \text{non-adjacent } \deg(u) + \deg(v) \geq n$



G has a **Hamilton circuit**



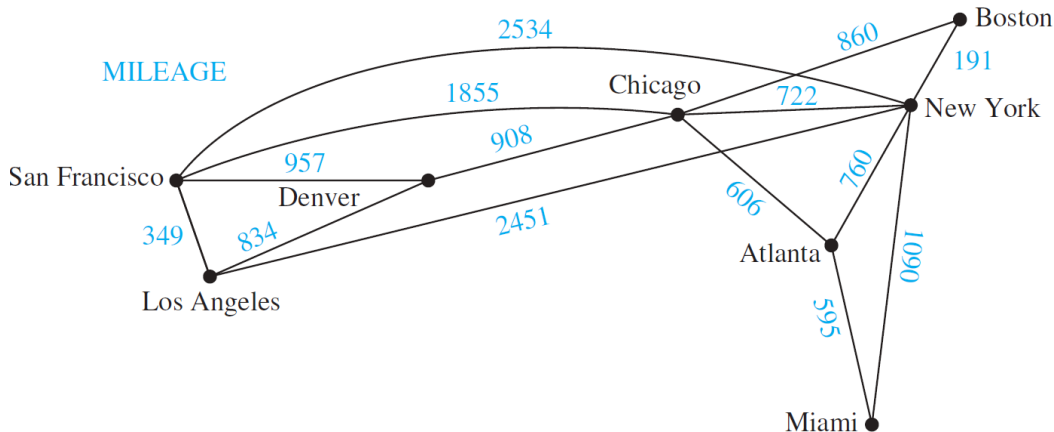
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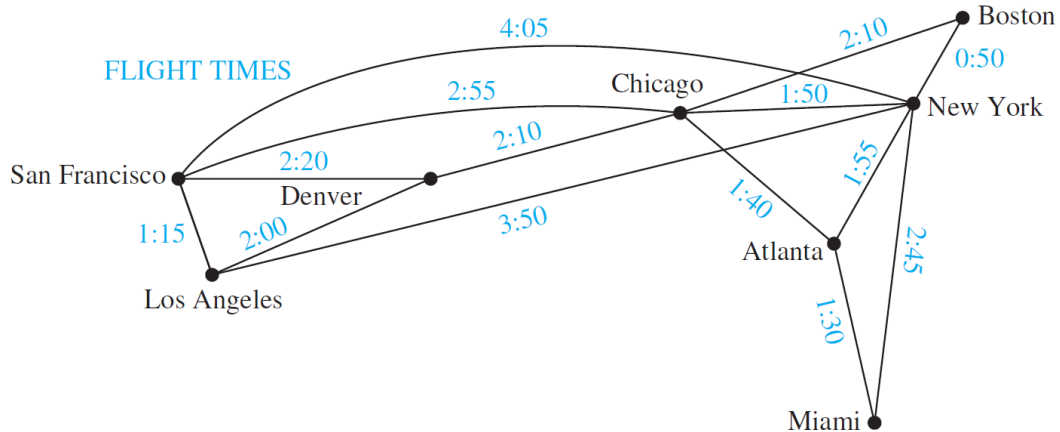
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7 Shortest-Path Problems



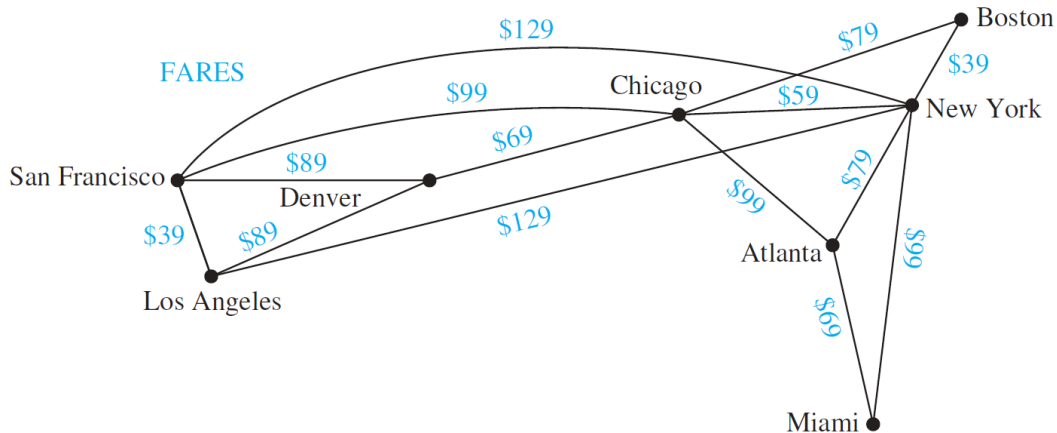
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7 Shortest-Path Problems



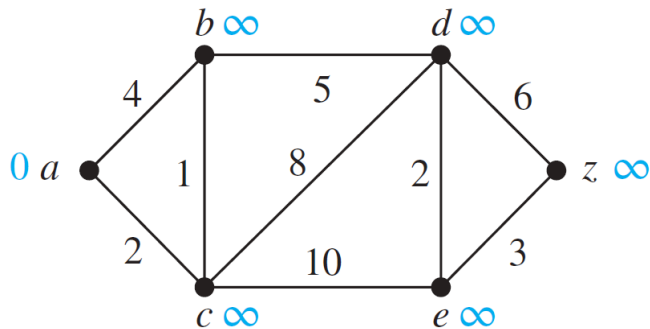
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7 Shortest-Path Problems



A weighted simple graph

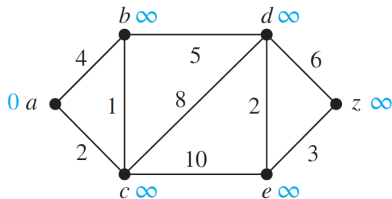
7 Shortest-Path Problems



Use Dijkstra's algorithm to find the length of a shortest path between the vertices a and z in the weighted graph displayed in Figure.

Example

7 Shortest-Path Problems

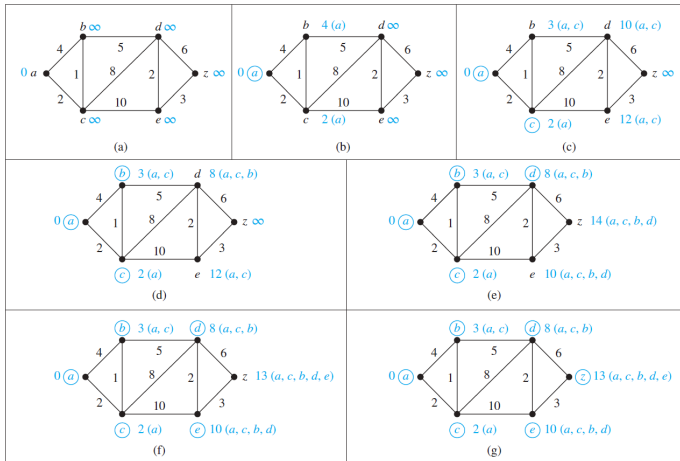


L(a)	L(b)	L(c)	L(d)	L(e)	L(z)	S
0	∞	∞	∞	∞	∞	$\emptyset$
	∞	∞	∞	∞	∞	$\{a\}$
	4	2	∞	∞	∞	$\{a, c\}$
	3		10	12	∞	$\{a, c, b\}$
			8	12	∞	$\{a, c, b, d\}$
				10	14	$\{a, c, b, d, e\}$
					13	$\{a, c, b, d, e, z\}$

$$z(13) \leftarrow e(10) \leftarrow d(8) \leftarrow b(3) \leftarrow c(2) \leftarrow a(0)$$

Solution

7 Shortest-Path Problems



A Shortest-Path Algorithm

7 Shortest-Path Problems

ALGORITHM 1 Dijkstra's Algorithm.

```

procedure Dijkstra( $G$ : weighted connected simple graph, with
    all weights positive)
    {  $G$  has vertices  $a = v_0, v_1, \dots, v_n = z$  and lengths  $w(v_i, v_j)$ 
      where  $w(v_i, v_j) = \infty$  if  $\{v_i, v_j\}$  is not an edge in  $G$  }
    for  $i := 1$  to  $n$ 
         $L(v_i) := \infty$ 
     $L(a) := 0$ 
     $S := \emptyset$ 
    { the labels are now initialized so that the label of  $a$  is 0 and all
      other labels are  $\infty$ , and  $S$  is the empty set }
    while  $z \notin S$ 
         $u :=$  a vertex not in  $S$  with  $L(u)$  minimal
         $S := S \cup \{u\}$ 
        for all vertices  $v$  not in  $S$ 
            if  $L(u) + w(u, v) < L(v)$  then  $L(v) := L(u) + w(u, v)$ 
            { this adds a vertex to  $S$  with minimal label and updates the
              labels of vertices not in  $S$  }
    return  $L(z)$  {  $L(z)$  = length of a shortest path from  $a$  to  $z$  }
  
```

Dijkstra's Algorithm

7 Shortest-Path Problems

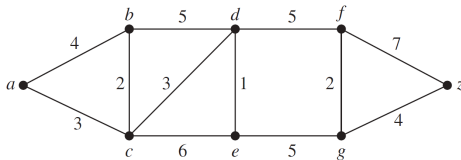
Theorem 7.1

Dijkstra's algorithm finds the length of a shortest path between two vertices in a connected simple undirected weighted graph.

Theorem 7.2

Dijkstra's algorithm uses $O(n^2)$ operations (additions and comparisons) to find the length of a shortest path between two vertices in a connected simple undirected weighted graph with n vertices.

Example 7.1. Find the length of a shortest path between a and z in the given weighted graph.



Dijkstra's Algorithm

7 Shortest-Path Problems

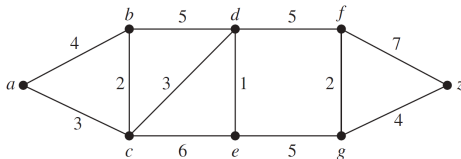
Theorem 7.3

Dijkstra's algorithm finds the length of a shortest path between two vertices in a connected simple undirected weighted graph.

Theorem 7.4

Dijkstra's algorithm uses $O(n^2)$ operations (additions and comparisons) to find the length of a shortest path between two vertices in a connected simple undirected weighted graph with n vertices.

Example 7.2. Find the length of a shortest path between a and z in the given weighted graph.



Solution. A shortest path between a and z is a, c, d, e, g, z and the length of the shortest path from a to z is 16.



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Quiz

8 Preview

Q1. Study a simple graph having the degree sequence $\{5, 5, 4, 4, 4, 3, 3, 2, 2, 2, 2, 1, 1\}$. If the graph has n edges, then $n = \dots$

Select one:

- a. 38
- b. 19
- c. 20
- d. 17

Q2. For which values of n do K_n have an Euler path but no Euler circuit?

Select one:

- a. 2
- b. 1
- c. 3
- d. $2m, \forall m > 1$

Q3. Every Euler circuit in K_{11} is a path of length ...

Select one:

- a. 45
- b. 22
- c. 11
- d. 55

Q4. If K_n has x edges and y vertices, then $x = \dots$, $y = \dots$

Select one:

- a. $n + 1$, n
- b. $2n$, $n + 1$
- c. $\frac{n(n + 1)}{2}$, n
- d. $\frac{n(n - 1)}{2}$, 2^n

Q5. The incidence matrix of the graph $K_{2,3}$ has the size of
Select one:

- a. 30
- b. 5×6
- c. 2×3
- d. 6×5

Q6. Study the statements:

- i. $K_{2,3}$ has no an Euler circuit, but has an Euler path.
- ii. $K_{2,3}$ has an Euler circuit.

Which statements is true?

Select one:

- a. None
- b. i
- c. ii
- d. Both

Answer. **Q1.** 19; **Q2.** 2; **Q3.** 55; **Q4.** $\frac{n(n-1)}{2}$, n ; **Q5.** 5×6 ; **Q6.** i



Q&A

Thank you for listening!